Chemical Engineering Journal 498 (2024) 155731

Contents lists available at ScienceDirect

Chemical Engineering Journal

journal homepage: www.elsevier.com/locate/cej

Study on synergistic effect of carrier combined with micro-aeration on anaerobic digestion of food waste

Keke Ding [a], Baocun Wu [a], Yi Wang [a], Linji Xu [a], Miao Liu [b], Jinjing Xiang [a], Yongdong Chen [a], Li Gu [a,*], Jinze Li [a,*], Lin Li [a], Qiang He [a], Shaowu Liu [c]

[a] Key laboratory of the Three Gorges Reservoir Region's Eco-environments, Ministry of Education, Institute of Environment and Ecology, Chongqing University, 174 Shapingba Road, Chongqing 400045, P.R. China
[b] Gastrointestinal Cancer Center, Chongqing University Cancer Hospital, 174 Shapingba Road, 400045, P.R. China
[c] Chongqing Water Group Co., Ltd, No. 1 Longjiawan, Yuzhong District, Chongqing 400043, P.R. China

ARTICLE INFO

Keywords:
Micro-aeration
Carrier
Food waste
Methane yield
Metagenomics

ABSTRACT

Enhancing the conversion efficiency of methane production from food waste (FW) is always a hot issue. Biological carriers and micro-aeration had demonstrated notable advantages in enhancing anaerobic digestion (AD) efficiency, but their synergistic potential remained largely untapped. This study introduced polyurethane carriers into micro-aeration assisted AD, trying to explore their synergism in enhancing the digestive efficiency of FW. Results revealed that an optimal blend of carriers and micro-aeration pose significant improvement to the digestive performance. Their combination enhanced microbial metabolic activity, accelerated the consumption or conversion of dissolved organic matter (DOM) molecules, promoted the aggregation of microorganisms, which improved their tolerance capacity under the oxygen stress brought by the micro-aeration, ensuring the system stability. It also enriched methane-producing microorganisms, and balanced methane-producing pathways. Additionally, the combination of micro-aeration and carriers promoted key enzyme expression in metabolic pathways, providing ample support for methane production. These findings held significance for optimizing FW treatment, improving methane production efficiency, and reducing the burden of organic waste treatment.

1. Introduction

In recent years, the management of organic waste, particularly food waste (FW), has become a global significant environmental challenge. It is predicted that 2.5 billion tonnes of FW are going to be generated by 2025 annually worldwide [1]. FW not only poses a threat to public health but also contributes to the emission of greenhouse gases, leading to climate change. Among the various available treatment methods, anaerobic digestion (AD) has emerged as a promising technology for efficiently converting FW into renewable energy and valuable by-products, with the process terminated by methanogenesis [2]. However, the problem of low organic matter conversion efficiency during AD has been consistently limiting its practical application [3,4]. Therefore, improving the conversion efficiency of FW organics into methane is a crucial concern.

Several studies have explored ways to improve the performance of AD, such as substrate pre-treatments [1], optimizing process parameters [5] and introducing additives [6]. However, the pretreatment or supplementary additives usually required extra energy and chemicals, which not only increased operational costs and complexity but also gave rise to a range of environmental issues. Recent studies have shown that micro-aeration, which involves injecting small amounts of air or oxygen (typically ranging from 0.005 to 5 L O_2/L_R-d) into an anaerobic system, with cost-effectiveness, has gained attention as a potential strategy for improving digestive performance due to its ability to accelerate the hydrolytic-acidification process, which provides more available substrate for methanogenesis [7,8]. Compared to strict anaerobic conditions, injecting trace amount of air creates a higher oxidation–reduction potential (ORP) environment that reshapes the microbial community in the digester. Facultative anaerobes involved in the hydrolytic-acidification process are stimulated and exhibit much higher activities. An appropriate micro-aeration contributes to the enhancement of the hydrolysis-acidification process and thus facilitates subsequent methanogenesis [9]. Ruan et al. [10] found that micro-aeration can enhance the microbial activity, increase substrate degradation rates, and improve overall process stability. Furthermore, according to Fu et al.'s research

* Corresponding authors.
E-mail addresses: guli@cqu.edu.cn (L. Gu), Lijinze@cqu.edu.cn (J. Li).

https://doi.org/10.1016/j.cej.2024.155731
Received 2 July 2024; Received in revised form 26 August 2024; Accepted 10 September 2024
Available online 12 September 2024
1385-8947/© 2024 Published by Elsevier B.V.

in digestion of corn stover, micro-aeration contributed to promote the biodiversity of facultative hydrolytic bacteria and acidifying bacteria [11]. Although the advantages of micro-aeration have been reported, precise control of oxygen dose is required when applying micro-aeration to substrates with high hydrolysis rates. Excessive aeration beyond the oxygen consumption and anti-oxidation capacities of facultative bacteria may have detrimental effects on obligate anaerobes, as high concentrations of Reactive Oxygen Species (ROS) can damage their phospholipid membranes and DNA [12].

Another key factor contributing to the poor performance of AD is the limited energy acquisition rate of anaerobic microorganisms, resulting in slow growth, particularly for methanogenic archaea [13,14]. In the AD of organic matters, daily slurry discharge poses a risk of expelling slow-growing microorganisms from the reactor, leading to an insufficient population of functional microorganisms. So, introduction of microbial attachment media is helpful for retaining functional anaerobes through formation of microbial agglomerations or biofilms [15]. Under the effect of the attachment media, here coexist of the attached biofilm and suspended sludge in the digester, which not only improves the number of microorganisms, but also effectively retains the slow-growing functional anaerobes involving in anaerobic conversion of organic pollutants [16]. Though some small particles such as zeolite, active carbon (AC), magnetite and et al, have been utilized in current researches, they pose insufficient surface for microorganism attach [17–19]. Besides the porous structures, biocompatibility is an important index for the media [20], which determines the efficiency of mass transfer between substrates and microbes [21]. Polyurethane is a synthetic polymer hydrophilic bio-carrier with interconnected pores, has multiple advantages because of its important properties like adsorption, bio-attachment, high porosity and contains some cationic surface active groups and hydrophilic groups such as hydroxyl, which enable firm fixation of microbes and enzymes onto the surface [22]. Wang et al.[23] discovered that the application of polyurethane carriers in digesters for municipal biowaste (FW, fruit-vegetable waste and dewatered sewage sludge) improved the digestive efficiency, and enhanced the microbial retention. It also contributed to high system stability due to the reduced mass-transport resistance and high microbe density of the biofilm. Shi et al.[24] have proved that adding polyurethane carriers to reactors increases microbial richness and diversity, benefiting their long-term stability. Additionally, the filling ratio of the attachment media also poses some impacts. A low filling ratio may result in poor substrate flow, while a high filling ratio can affect substrate diffusion and metabolites [25].

Considering the effectiveness of both bio-carriers and micro-aeration in enhancing AD, their combination may demonstrate even greater potential. Biofilms can mitigate the toxic effects of micro-aeration on obligate anaerobes [26,27]. Additionally, biofilms can effectively trap slow-growing functional microorganisms (e.g., methanogenic archaea), while facultative bacteria with higher energy metabolism through aerobic respiration can grow in suspension, utilizing enhanced biological activity to improve their ability to hydrolyze and acidify organic matters [24,28]. Complex organic compounds are hydrolyzed into dissolved organic matter (DOM) during the hydrolysis and acidification stage, and DOM is further fermented into volatile fatty acids (VFA) [29], providing metabolizable substrates for slow-growing archaea attached to the surface of the carrier, thereby enhancing overall treatment efficiency. DOM serves as the primary substrate for VFA production by fermentation bacteria. Many studies have shown that the characteristics and variations of DOM significantly impact the final utilization of DOM products, with changes in VFA and biogas production closely linked to variations in DOM characteristics [30–32]. However, it is currently unclear whether carrier and micro-aeration affects the characteristics of sludge DOM molecular composition, and how the key DOM components are transformed and utilized in methanogenesis. Traditional spectroscopic analysis methods can only characterize the DOM composition at a macro level, making it difficult to reveal the DOM composition features and their relationship with AD at the molecular level. Fourier transform ion cyclotron resonance mass spectrometry (FT-ICR-MS), with its ultra-high resolution (>200,000), mass accuracy (error < 0.000001), and sensitivity, can completely separate complex organic molecular mass spectrometry peaks in DOM, thus enabling the resolution of DOM composition features at the molecular level [33].

The above analysis has shown that both bio-carriers and micro-aeration can effectively promote AD. However, research on their combined application in this process remains limited, particularly within solid waste treatment. Therefore, this paper tries to combine bio-carriers and micro-aeration to enhance AD performance of FW. The study first analyzed the operational performance of the reactor to evaluate their promotion effects and obtain the optimal operating conditions. Then, carrier morphology, enzyme activity, and DOM characteristics under these conditions were compared. Finally, metagenomic sequencing was used to explore the reactor's metabolic pathways, revealing underlying microbial regulatory mechanisms. The results of this study provide new insights for further enhancing the digestive performance of FW.

2. Materials and methods

2.1. Digester configuration and operational conditions

The anaerobic digester was made of a plexiglass cylinder (inner diameter 8 cm, height 25 cm) with an effective working volume of 0.7 L, including a feed hopper, a mixing shaft, a separate jacketed thermostat and gas collection device (Fig. S1). The polyurethane foam carriers were purchased from Henan Tianyi Environmental Protection Technology Co., Ltd. Detailed information can be found in Table S1, and the actual photo is also presented in the supporting information (Fig. S2). Experiments were conducted with different filling ratios ($V_{carrier}/V_{reactor}$) of 0 % (D0), 10 % (D10), 20 % (D20), and 30 % (D30). The AD systems were operated in semi-continuous feeding mode to maintain a constant hydraulic retention time (HRT) of 20 d, with an organic loading rate (OLR) of 2.5 g VS/L·d, and the digestion temperature was maintained at 35 ± 1 ℃ by a thermostat water bath (HH-501, Xinbao, China). At the end of each discharging-feeding, the reactors were flushed with N_2 gas for 5 min to remove the air.

Anaerobic inoculum was obtained from Tangjiatuo Wastewater Treatment Plant located in Chongqing, China. The inoculum was settled naturally under anaerobic conditions for 1 day to remove the supernatant, and then the resulting sludge was filtered through a 20-mesh sieve to remove large impurities. The prepared inoculum was stored at anaerobic conditions of 35 ± 1 ℃ for more than a week before use to eliminate the interference of residual biodegradable organic matter to subsequent AD process. The substrate FW was artificially prepared in the laboratory following the formula stated in [34]. The details of FW and inoculum are listed in Table 1. The digester was run continuously for a total of 85 days and was divided into two stages according to whether micro-aeration was introduced, namely the without micro-aeration stage (day 1–35) and the micro-aeration stage (day 36–80). During the second stage, a uniform and gradual introduction of air was achieved by

Table 1
Properties of FW and inoculum.

Parameter	Food waste	Inoculum
TS (g/L)	126.9 ± 0.6	64.6 ± 0.4
VS (g/L)	123.1 ± 0.4	22.1 ± 0.5
VS/TS (%)	96.9 ± 0.5	34.2 ± 0.4
pH	4.9 ± 0.1	7.5 ± 0.2
TCOD (mg/L)	167215 ± 4260	38045 ± 2410
SCOD (mg/L)	51637 ± 1390	1716 ± 320
Crude fat (g/L)	19.9 ± 0.5	NA
Crude protein (g/L)	18.4 ± 1.7	NA
Carbohydrate (g/L)	71.4 ± 4.0	NA

*Data shown as mean ± standard deviation of triplicate samples. NA: Not available.

injecting 5 mL using a syringe after each feeding, resulting in an increase in ORP by 30–50 mV.

2.2. Chemical and physical analysis

pH was determined by pH meter (FE28, METTLER TOLEDO, Switzerland). Measurements of total solid (TS), volatile solid (VS) and chemical oxygen demand (COD) were conducted in accordance with standard methods [35]. Carbohydrates, crude protein and crude fat were quantified according to [36]. The biogas composition and volatile fatty acids (VFAs) were determined by a gas chromatography (GC1690, Jiedao, Hangzhou). The operation process of gas chromatography was introduced in detail in our previous study [37]. The extraction method of EPS refers to the study of [38]. The PN and PS of each form of EPS were determined as the total amount by Lowry-Folin method and phenol–sulfuric acid method, respectively [39]. α-Glucosidase activity, ATP content, acetate kinase, ATP and coenzyme F_{420} were measured by using the ELISA kit (Shanghai hengyuan biological technology co., LTD). The morphology of the carrier and suspended sludge was observed using a fluorescence microscope (SEM, Hitachi S-4800, Leica, Germany). Spatial microstructure of suspended sludge was visualized by CLSM (Leica, TCS SP8 STED 3X, Germany). Detailed staining and observation were performed according to the previously reported method by [40].

2.3. FT-ICR-MS analyses

Supernatants were separated from the samples of digested sludge by centrifuge at 6000 rpm for 20 min, and filtered through 0.45 μm membrane. The filtered supernatant was further diluted to TOC=2 ~ 3 mg/L with an ultrapure water to avoid protein precipitation in DOM [41]. Solid phase extraction (SPE) was performed using PPL cartridges (BondElute PPL, 6 mL, 500 mg, Agilent Technologies, USA) after determination of DOC in the supernatants. The molecular composition of DOM was characterized using Bruker SolariX FT-ICR-MS (15.0 T superconducting magnet) equipped with electrospray ionization (ESI) ion source. Samples for ESI FT-ICR-MS analysis were continuously infused into the ESI unit by syringe infusion at a flow rate of 120 μL/h. The ESI needle voltage was set to − 4.0 kV, and the ion accumulation time was set to 0.2 s, mass range of 100–1600 Da. A total of 300 continuous 4 M data FT-ICR transients were added to enhance the signal-to-noise ratio and the dynamic range. The range of m/z was set from 100 to 800. A sodium formate solution (10 mM) was used to externally calibrate the instrument, and the known series of natural organic matter (NOM) was used to internally recalibrate the spectra. The detailed operation is described in the Supporting Information.

2.4. Metagenomic sequencing and analysis

The anaerobic biomass samples were collected the reactors at the end of Stage II and III and stored in −80 °C for microbial analysis. The DNA extraction of the samples was performed using the E.Z.N.A.® Soil DNA Kit (Omega Bio-tek, USA). After the extraction of genomic DNA, the concentration and purity of DNA were detected using TBS-380 and NanoDrop2000 respectively, and the integrity of DNA was determined by 1 % agarose gel electrophoresis. Fragmentation of DNA was achieved using Covaris M220 (Gene Company, China) to obtain fragments of approximately 400 bp. The PE library was constructed using the NEXTFLEX Rapid DNA-Seq kit (Bioo Scientific, USA). The bridge PCR amplification was followed by sequencing on the Illumina NovaSeq/Hiseq Xten platform (Illumina, USA) for metagenomic sequencing.

The raw data was quality-controlled using the Fastp software. The reads were aligned to the host DNA sequences using the BWA software, and contaminant reads with high similarity were removed. The optimized sequences were assembled using the MEGAHIT software based on the principle of succinct de Bruijn graphs. Contigs with a length of ≥ 300 bp were selected from the assembly results as the final assembly. The contigs were then subjected to ORF prediction using MetaGene, and the gene sequences predicted from all samples were clustered using the CD-HIT software to construct a non-redundant gene set. Finally, the SOAPaligner software was used to align the high-quality reads of each sample to the non-redundant gene set (with 95 % identity) and to calculate the abundance of genes in the corresponding samples.

2.5. Statistical analysis

The experimental data underwent analysis of variance (ANOVA) for statistical processing. All physicochemical index data presented in this paper are expressed as "mean ± standard deviation" (n = 3) to improve clarity. *Student's t* test was utilized to determine whether significant differences existed between different systems, with a p-value below 0.05 considered statistically significant. Statistical mapping was conducted using Origin 9.0 software.

3. Results and discussion

3.1. Digestive performance

Fig. 1 shows the digestive performance of different digesters. It was obvious that the presence of carriers increased methane production (Stage I). The averaged specific methane yield for D20, D30, and D10 are 390.7, 381.6, and 350.6 mL CH_4/g·VS, which are much higher than that of D0 (about 330.1 mL CH_4/g·VS). The specific methane production in the D30 reactor is lower than that in D20, possibly due to the excessive input of carrier, affecting the liquid-phase mass transfer in the digester.

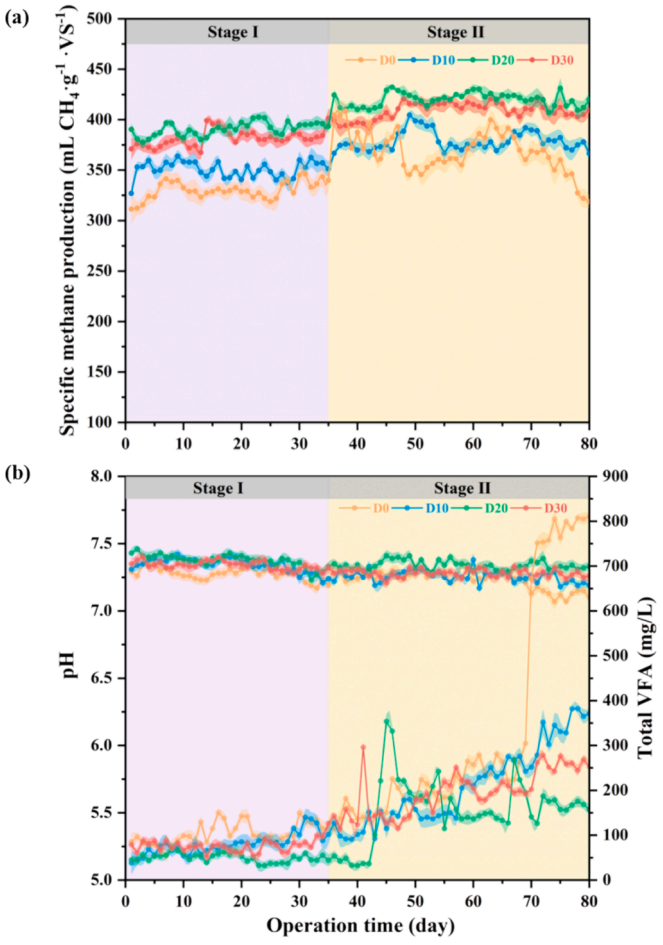


Fig. 1. (a) Specific methane production and (b) TVFA and pH in different digesters.

Previous studies have pointed out that a high filling rate of carriers can lead to an increase in biofilm thickness, affecting the supply of substrates and the discharge of metabolites within the internal biological phase, thereby increasing mass transfer resistance [42].

After introducing micro-aeration in stage II, the methane production for all digesters got increased, indicating that the micro-aeration has a positive effect on methane production. As can be seen from the figure, the micro-aeration treatment resulted in a slight improvement in methane production in the control group (D0), with a specific methane yield of 368.84 mL CH_4/g·VS. Zhou et al.'s study also indicated that the promoting effect of low-dose oxygen on methane production in AD was not significant [43]. With the proceeding of the micro-aeration, there is a rapid increase in VFA concentration in D0, resulted in declining in the methane yield. However, when carrier was present in the digester, the introduction of micro-aeration gained more promotion of the methane production, particularly for D20 (417.7 mL CH_4/g·VS), reaching 1.13 times that of the control. Compared the system failure at the end of the experiment in the control digester, the participation of the carrier ensured the stability of the digester without declining of the methane production.

As important intermediate compounds in methane production metabolism, VFAs' concentration is a crucial consideration for the optimal performance of biogas digesters. Therefore, it is necessary to monitor VFA concentrations to study the stability and efficiency of biogas digesters. Fig. 2b depicts the TVFAs' concentration throughout the experiment. In stage I, the TVFAs concentration in the carrier groups remained relatively stable. In contrast, the TVFAs concentration in the control group (D0) fluctuated greatly. The polyurethane carrier provided attachment and growth space for microbial cells, reducing the risk of microbial loss and increasing the stability of the anaerobic digestion system [23]. As the reactor operated and entered the early period of stage II, the control group (D0) exhibited relatively stable TVFA concentrations and pH values. However, with the continuous introduction of micro-aeration, the TVFAs' concentration increased gradually to as high as 800 mg/L, while the pH value slowly declined to around 7.1, indicating the occurrence of acidification within the system. This may be attributed to the generation of ROS during the reaction process, which could have exerted a certain toxic effect on obligate anaerobes such as methanogenic archaea [44]. Compared to the control digester (D0), the TVFA in the carrier participated groups maintained at much lower concentration levels, ensuring the stable pH throughout the experiment. It is also noticed from the figure that there was a slight accumulation of TVFA in D30 and D20 during the early stage after micro-aeration introducing. This can be ascribed that the archaea were still in the adaptation period to the new environment, posing lower acids consumption rate to the acid generation in the hydrolytic-acidification process [45]. The microbial community gradually adapts to the micro-aerobic environment, and thus the system maintained stable throughout the experimental period [46].

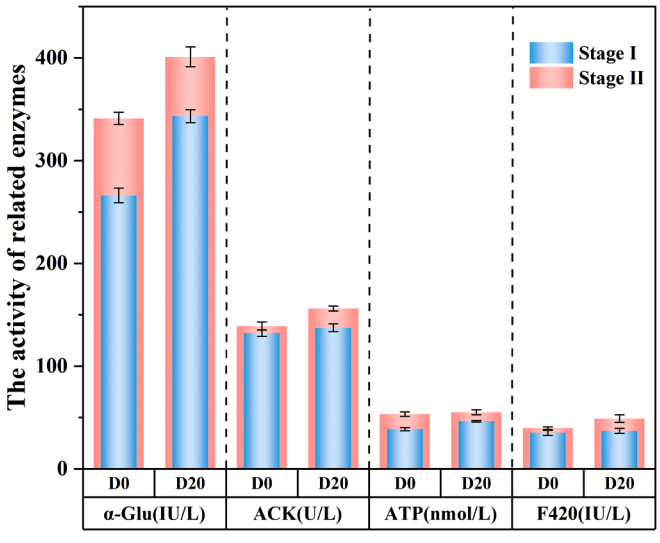


Fig. 2. Enzyme activity before and after micro-aeration.

3.2. Microbial metabolic activity

Considering the highest methane production efficiency observed in D20, it was selected for in-depth analysis in the subsequent study. AD is a biological process catalyzed by microorganisms through the secretion of specific enzymes, and the enzymes activities reflects the metabolic activities of microorganisms. Therefore, enzyme activities in different digesters were monitored and compared (Fig. 2). α-glucosidase is the main extracellular hydrolytic enzyme in anaerobic metabolism [47]. Without micro-aeration, the activity of α-glucosidase in D20 is 128 % of that in the control group (D0), verifying the positive effect of the carrier. Further enhancement of α-glucosidase activity was observed in D0 and D20 when micro-aeration was introduced into the digester. Similar to α-glucosidase, micro-aeration also led to an increase in the activity of acetate kinase (ACK), especially in the digester containing the carrier. The increased activities of α-glucosidase and ACK verified that the micro-aeration enhanced the hydrolytic-acidification process [34]. The carrier provided a habitat for functional bacteria responsible for hydrolysis and acidification, promoting better interactions among bacteria, enzymes, and substrates to facilitate the conversation of organic matters [48]. In addition, the ATP content in the system also exhibited a significant increase during the stage II, approximately a 37 % and 18 % rise in ATP levels for D0 and D20. Previous studies have demonstrated that the addition of trace oxygen as a terminal electron acceptor in anaerobic bioreactors can accelerate the hydrolysis of complex organic compounds by facultative bacteria and enhance the electron transfer activity in microbial metabolism [49]. The reduction of O_2 by NADH (Nicotinamide Adenine Dinucleotide) releases a substantial amount of energy, leading to the generation of a proton gradient that drives ATP synthesis and the transport of other metabolites, thereby increasing the overall system's productivity and metabolic efficiency [50].

Methane production during AD determines heavily by the activities of methanogenic archaea. The presence of coenzyme F_{420}, acting as a distinct electron carrier, can reflect the activity of methanogenic archaea [51]. As shown in Fig. 2, the coenzyme F_{420} content in digesters D0 and D20 in stage I was 35.07 and 36.98 IU/L. After the introduction of micro-aeration, the activity of F_{420} increased in both digesters. These findings indicated that micro-aeration can enhance the secretion of F_{420} by microorganisms [3]. Furthermore, in the digesters with carriers, the coenzyme F_{420} content was higher than in the control. This reveals that the micro-oxygen environment conducive to archaea's metabolic activity [47], by potentially inducing their aggregation to form biofilm on the carrier to cope with oxygen stress [52].

3.3. EPS content and spatial distribution of biopolymers

Extracellular polymeric substances (EPS) secreted by microorganisms were a main component in biofilms, which can embed microbial cells in a three-dimensional matrix within the floc, playing an important role in maintaining the stability of the sludge floc and protecting microbial cells [30,53]. In Fig. 3a, EPS were classified as loosely bound (L-EPS) and tightly bound (T-EPS) ones based on their degree of combination to the cell surface, with their composition characterized by protein and polysaccharide concentrations. In the absence of micro-aeration, higher levels of various EPS were observed in the carrier group compared to the control group. This observation confirms that the carrier created a favorable environment for microbial proliferation and attachment, thereby promoting microbial growth and EPS production. After micro-aeration was implemented, a significant increase was observed in both the L-EPS and T-EPS of the control group. Surprisingly,

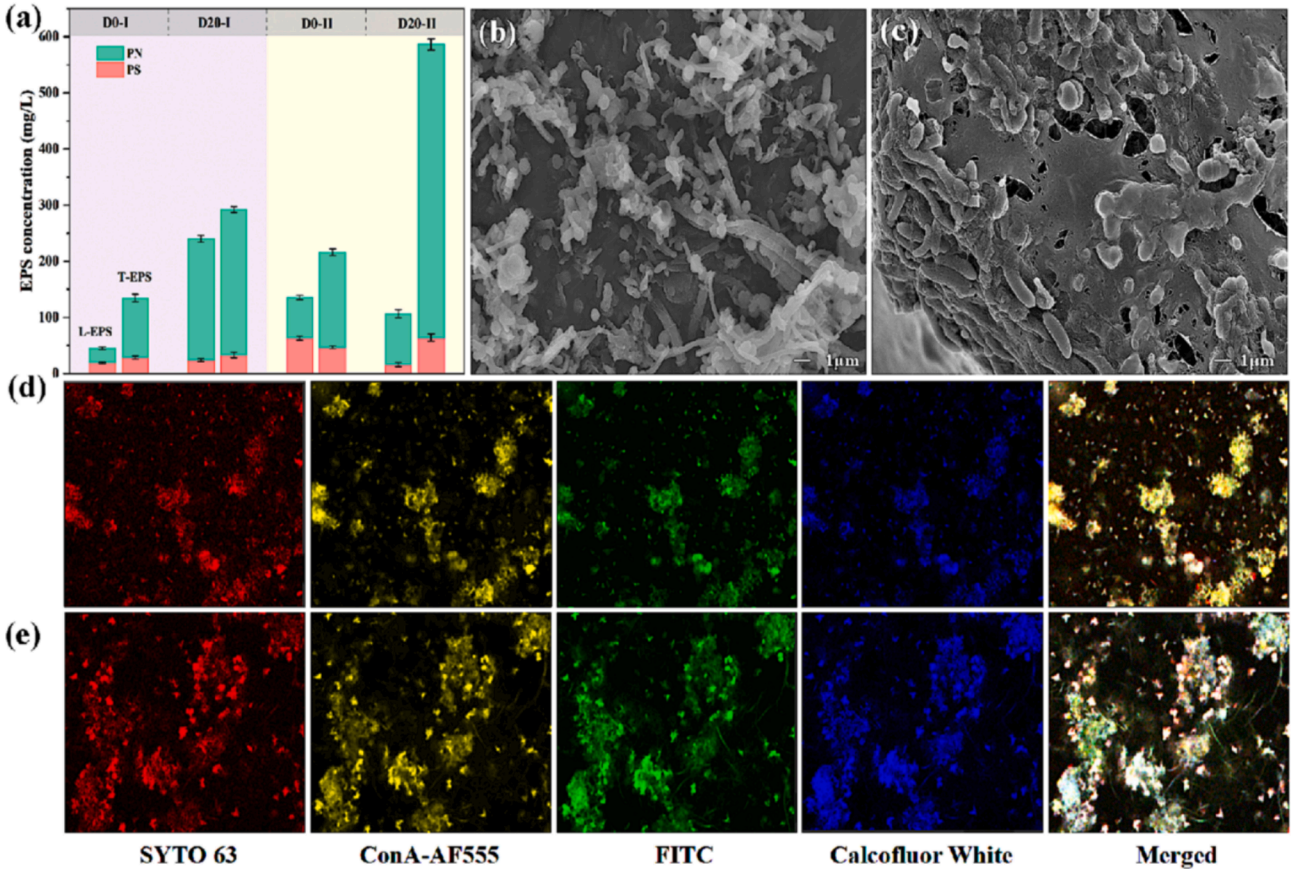


Fig. 3. (a) EPS content at different operational conditions; SEM analysis of anaerobic biomass in the micro-aeration stage: (b) D0 and (c) D20 carriers; CLSM images of sludge in the micro-aeration stage: (d) D0 and (e) D20.

the distribution of different EPS components in the D20 digester underwent substantial changes. The L-EPS content decreased while the T-EPS content showed a steep increase, surpassing that of L-EPS by a large margin. T-EPS possesses a distinct shape and forms a tight, stable attachment to the cell surface, contributing to the improved stability of the system during micro-aeration operation [54]. In terms of composition, it was evident that PN constituted a high proportion, while the content of PS was minimal and almost negligible. Earlier research had indicated that hydrophobic proteins in EPS facilitated microbial aggregation and promoted biofilm formation [55]. This effect could be attributed to the abundance of negatively charged amino acids in proteins, making them more prone to electrostatic binding with multivalent cations-a crucial factor in stabilizing aggregate structures [56].

To further visualize the attachment behavior of key microorganisms during the micro-aeration stage, SEM analysis was conducted on the sludge in D0 and the carrier in D20, to reveal their differences in microbial morphology. The surface of the sludge appeared coarse with dispersed microbial distribution in D0 (Fig. 3b). In contrast, the microorganisms in D20 carriers exhibited a denser structure, with plump and clearly distinguishable microbial cells. A layer of biofilm was deposited on the surface of microorganisms, with rod-shaped and spherical microorganisms distributed and embedded within it (Fig. 3c). Ultimately, anaerobic microorganisms were almost enveloped by the macromolecular metabolites to protect them from the toxicity of ROS. Moreover, the spatial distribution of organic biopolymers (including nucleic acids (red), α-D-glucopyranose polysaccharides (yellow), proteins (green), and β-D-glucopyranose polysaccharides (blue)) in the sludge after micro-aeration treatment was observed using CLSM. SYTO 63, Con A-AF555, FITC, and calcofluor white fluorescent substances were respectively used for labeling, with Merged representing the combined images of the four dyes. The staining results revealed that the sludge was surrounded by PN and PS, which were considered as the primary contributors to biofilm formation [57]. Proteins and β-D-glucopyranose polysaccharides were primarily concentrated within the core of aggregates, while nucleic acids and α-D-glucopyranose polysaccharides were positioned at the periphery (Fig. 3d and 3e). This distribution pattern suggested a possible correlation with EPS, consistent with the findings of regarding EPS and cell distribution in granules [40]. The network structure of PS molecules promoted sludge flocculation, with β-PS as the backbone of the network playing a significant role. α-PS, PN, lipids, and cells easily associated with β-PS to maintain the structural stability of flocs and improve sludge bio-flocculation accordingly [58]. Notably, the D20 sample exhibited heightened protein secretion by cells, with a significant presence of filamentous microorganisms and a denser cell distribution, tending to aggregate into clusters (Fig. 4e). After micro-aeration, the increase in protein content of EPS in digester D20 enhanced the physical bridging of neighboring bacterial cells and formed a cross-linked network by binding with polyvalent cations and organic molecules [59] (Fig. 3a). This phenomenon might be linked to microorganisms creating a denser EPS structure for self-preservation during stressful environmental conditions.

3.4. DOM molecular characteristics

3.4.1. General characteristics of sludge DOM

Dissolved organic matter (DOM) are believed to be a carbon source that microorganisms can directly assimilate during AD [60]. Understanding the transformation of organic components in AD can be greatly facilitated by identifying the characteristics of DOM. In this study, FT-ICR-MS was used to identify 4571,5859,4808 and 3397 molecules in D0-I, D20-I, D0-II and D20-II, respectively, with detailed molecular characteristics summarized in Table 2. The molecular weight

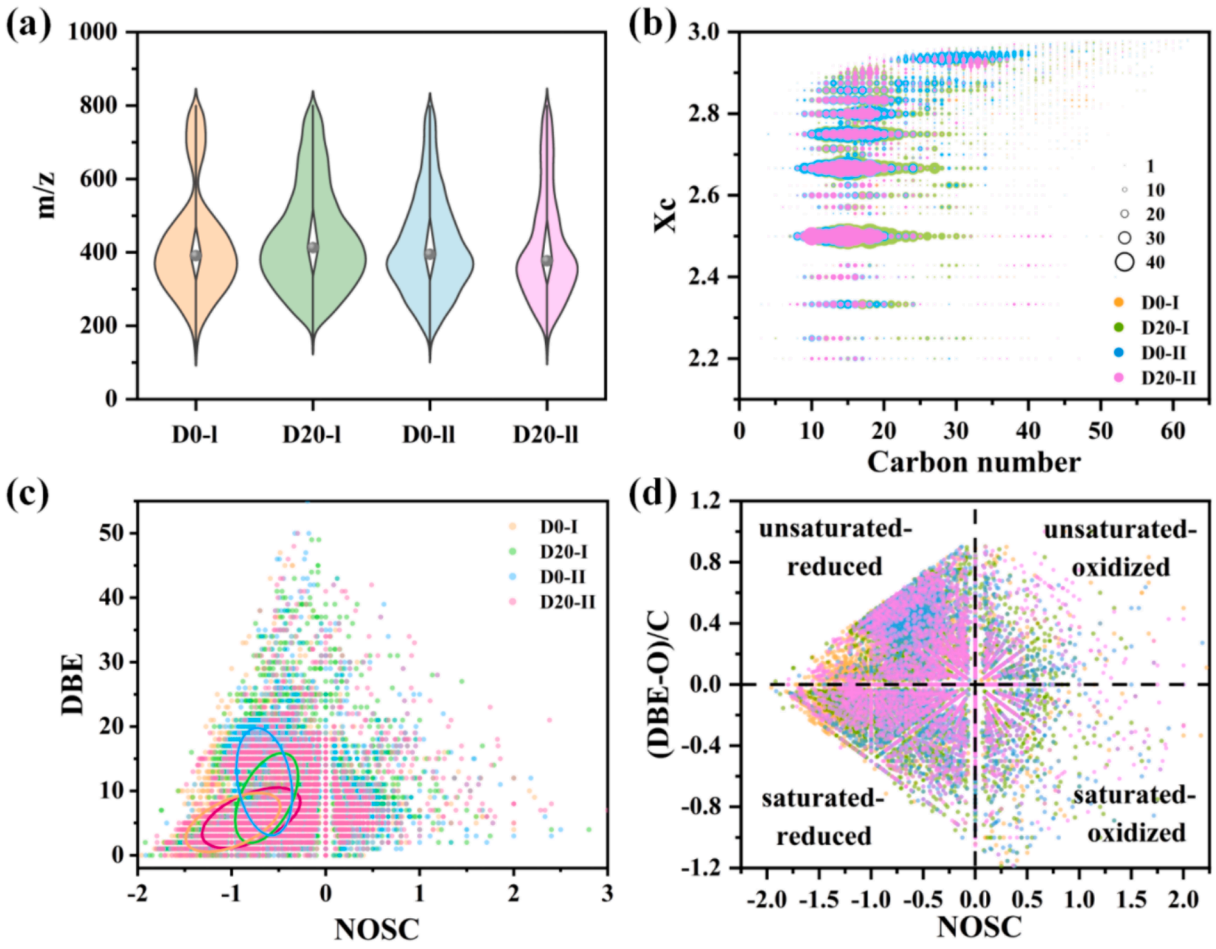


Fig. 4. (a) Violin plots of molecular mass distribution in D0 and D20 under different operational conditions; (b) A Plot of Xc versus carbon number; (c) NOSC and DBE of molecules; (d) (DBE-O)/C-NOSC plots of sludge in D0 and D20.

Table 2
Molecular characteristics under different operational conditions.

Parameters	Without micro-aeration		Micro-aeration	
	D0	D20	D0	D20
M_w	562.15	476.55	427.53	455.76
C_w	30.73	25.19	21.98	24.33
H_w	53.42	31.90	26.75	29.97
O_w	6.42	7.50	7.18	6.80
N_w	1.41	0.83	0.92	0.95
S_w	0.00	0.33	0.28	0.36
H/C_w	1.76	1.37	1.34	1.31
O/C_w	0.23	0.34	0.38	0.33
DBE_w	5.23	10.32	9.57	10.16
$NOSC_w$	−1.13	−0.54	−0.41	−0.46
DBE/C_w	0.18	0.38	0.41	0.42
AI_{modw}	−0.01	0.24	0.27	0.20
CHO (%)	31.44	27.70	29.68	27.32
CHON (%)	36.21	41.90	38.83	44.04
CHOS (%)	17.39	19.24	22.03	15.63
CHONS (%)	14.96	11.16	9.46	13.01
Lipids (%)	19.38	6.78	3.89	9.45
Proteins (%)	27.11	27.58	24.06	23.46
Carbohydrates (%)	2.08	5.14	6.41	5.80
Unsaturated Hydrocarbons (%)	4.68	10.80	11.75	7.30
Lignins (%)	39.49	40.23	44.22	42.51
Tannins (%)	1.27	3.16	2.48	2.06
Condensed Aromatic (%)	1.86	4.63	5.95	7.04
Others (%)	4.13	1.69	1.25	2.38
Sum	4571	5859	4808	3397

distribution of the compounds was monitored between 100–800 *m/z*, with relatively abundant compounds distributed in the range of 300–500 *m/z*, accounting for around 55 % of the total (Fig. S3a). The double bond equivalents (DBE) represent the total number of double bonds and rings in a molecule, and can be used to characterize degree of unsaturation or complexity of the molecule. A higher DBE value signifies a wider variety of compounds with an increased number of double bonds and rings. The nominal oxidation state of carbon (NOSC) is related to the biogeochemical reactivity and bioavailability of organic compounds, with organic molecules possessing high NOSC values being more thermodynamically favorable for degradation [61,62]. In the initial stage, DOM in D20 exhibited higher DBE and NOSC values (Fig. S3), potentially attributable to the addition of the carrier resulted in higher reactivity in biogeochemical reactions. Conversely, the DOM in D0 exhibited relatively lower values, indicating the presence of recalcitrant stable compounds that were difficult to be degraded [62].

Analysis of the molecular weight distribution (Fig. 4a) and changes in aromaticity equivalent (Fig. 4b) reveals that D20 exhibited a lower molecular range and decreased aromaticity after micro-aeration treatment, corresponding to its lower M_w and AI_w values. Both DBE and NOSC in D20 were shifted towards lower values (Fig. S3). Kellerman et al.[63] showed that intrinsic molecular properties were crucial for the overall reactivity of organic matter. Specifically, degradation processes preferentially removed DOM characterized by highly oxidized states and aromatic compounds, while reduced and aliphatic compounds exhibited resistance to degradation or underwent tight cycling. Consequently, the primary constituents in D20 were reduced (low NOSC) and aliphatic (low DBE) compounds after micro-aeration treatment (Fig. 4c).

Additionally, it was found that the DOM molecules in D20 exhibited lower O/C_w and DBE_w, while DOM molecules in D0 showed the opposite trend, indicating that large DOM molecules in D20 sludge are tend to be hydrolyzed into smaller organic molecules with higher saturation, lower oxygen content, and fewer aromatic structures under the effect of micro-aeration. Studies have shown that such organic compounds typically represent more biodegradable components readily be utilized by microorganisms during AD [60]. Although DBE has been widely used to characterize complex samples, its drawback is that compounds with heteroatoms (e.g., O, S) can overlap in terms of DBE. For instance, oxygen and sulfur don't affect the DBE value, but they may add to the potential double bonds [64]. Therefore, to estimate the number of C–C double bonds in AD mixtures, (DBE-O)/C was utilized to further visualize the changes in DOM induced by carriers and micro-aeration. Based on the positive and negative values of (DBE-O)/C and NOSC, the sludge DOM molecules were classified into four categories (Table S2). A higher positive value of (DBE-O)/C indicated increased unsaturation, while a higher positive NOSC value signified greater oxidation [65]. The FW DOM molecules were predominantly in a reduced state, with more unsaturated species than saturated species, and fewer molecules in the oxidized state region (Fig. 4d), indicating that unsaturated and reduced compounds were preferentially decomposed by the enhancement of micro-aeration [66,67].

3.4.2. *Composition and transformation of DOM*

The above analysis revealed significant differences in the general characteristics of DOM molecules in D0 and D20 before and after micro-aeration treatment, prompting further investigation into whether micro-aeration affected the molecular composition of DOM in the digested organics. By utilizing the different H/C and O/C characteristics of typical compounds, the Van Krevelen diagram of DOM molecules was divided into seven categories of typical compounds, demonstrating the distribution of compounds and differences in molecular composition among DOM samples [68] (Fig. S4). Based on the elemental composition of DOM molecules, the DOM molecules in sludge were classified into four types (CHO, CHON, CHOS, and CHONS). Among them, CHO and CHON compounds were the predominant constituents of DOMs in sludge, with CHON compounds representing the highest proportion, ranging from 36.21 %~44.04 % of the total DOM molecules (Table 2). This suggested that the released DOM molecules mainly comprised

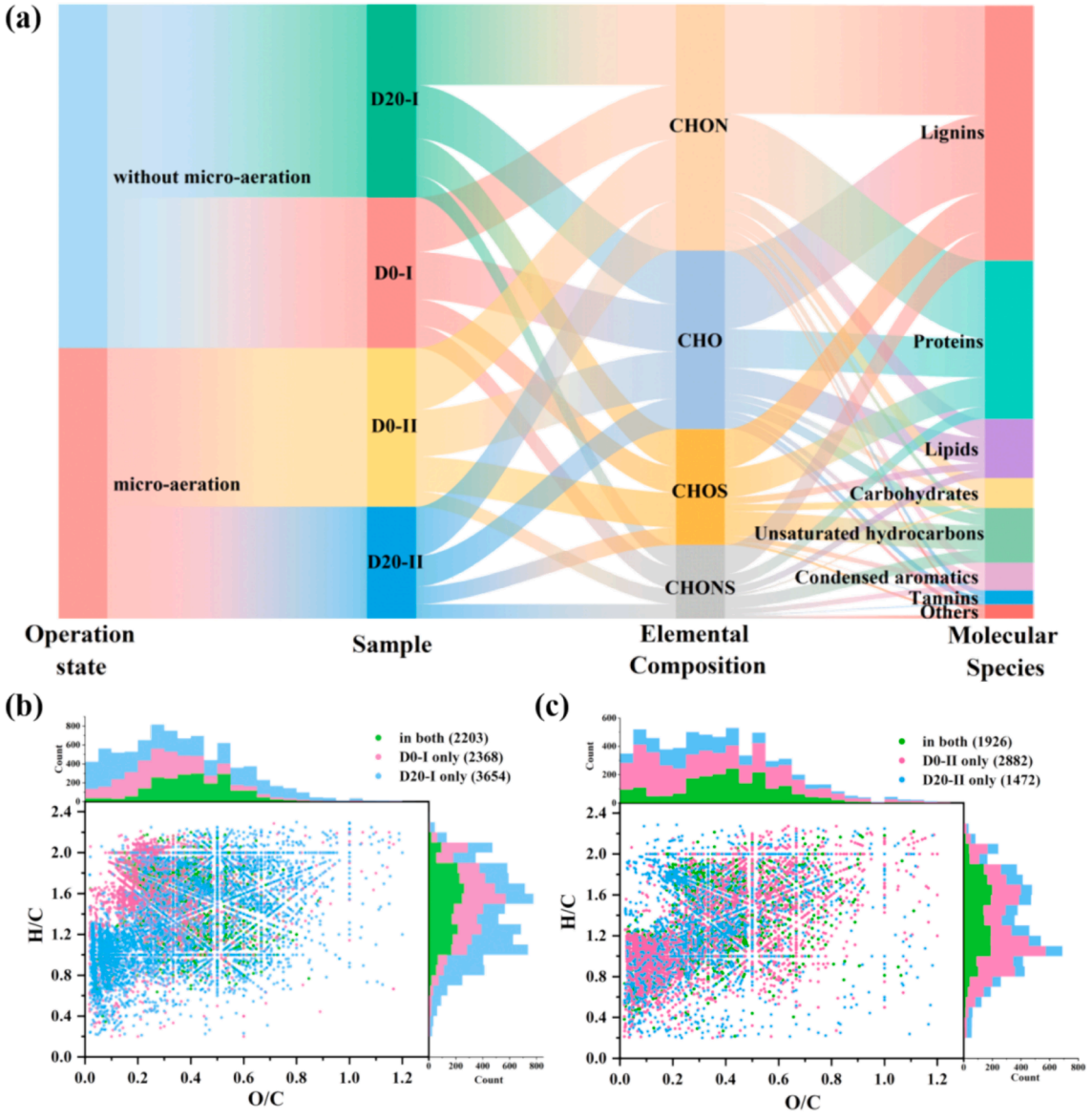


Fig. 5. (a) The distribution diagrams of elemental composition and molecular species of DOM_{D0} and DOM_{D20}; Differences in molecular composition of D0 and D20 (b) before and (c) after micro-aeration treatment.

nitrogen-containing organic compounds, which was related to proteins being the main component of organic matter in the substrate [33]. After undergoing AD process assisted by micro-aeration, the composition of DOM molecules exhibited changes. Specifically, there was an increase in the presence of CHON compounds in both D0 and D20 samples, with a higher content of CHON compounds observed in the DOMs of the carrier group compared to the control group [69]. Furthermore, a decrease in the proportion of CHONS compounds was noted in the D0 samples, while the abundance of CHOS compounds within the DOM molecules showed an increase (Fig. 5a, Table 2). Surprisingly, opposite trends were observed for these compound types in the D20 samples, indicating that the AD system incorporating the carrier could effectively treat organic compounds with N- and S-containing functional groups, and the stage III demonstrated superior efficacy compared to the stage II in treating these substances, primarily due to the higher dissolved oxygen accelerating the biodegradation [70]. Recent studies have also highlighted the ability of certain organic compounds (e.g. cysteine, serine, sulfur-containing amino acids), although difficult to be degraded in AD themselves, can promote the degradation of VFAs by decreasing oxidation–reduction potential, mediating electron transfer, and enhancing microbial metabolic activity, thereby improving methane production [71,72].

In Fig. 5a, lignins were observed to be the predominant compound type in all digesters. The proportion of lipid compounds in group D0 was significantly higher than in D20, indicating the poor biodegradability of lipids pre-existing in FW or/and generated through microbial activities in D0 [73]. Compared to the control group, adding bio-carriers increased unstable compound abundance (e.g., proteins, carbohydrates and unsaturated hydrocarbons). After micro-aeration introduction, lipid and protein compounds in the control group significantly decreased, indicating that the micro-aeration promoted the degradation of lipid and protein substances, which were key precursors for bacteria to produce VFAs [74]. Protein and unsaturated hydrocarbon levels decreased in D20, while lignin and condensed aromatic levels increased. Continuous degradation of proteins and unsaturated hydrocarbons enriched aromatic structures during AD. Studies have shown that the decomposition of protein macromolecules induced by hydroxyl groups contributes to the enrichment of condensed aromatic compounds in sludge EPS [75]. Lignin-derived compounds are a typical type of recalcitrant DOM, containing reducible groups such as alkoxy or hydroxy substituents conjugated to aromatic rings via ether-linkages [33]. The increase in lignin-derived DOM may be due to the degradation of carboxylic and alicyclic compounds, which were components of many microbial cell envelopes and natural metabolites [76].

In addition, visualization was conducted for the shared and unique molecules in D0 and D20 to further elucidate the transformation of DOM molecules before and after the micro-aeration treatment (Fig. 5b and 5c). In stage I, there were 2203 shared DOM molecules in D0 and D20, primarily concentrated in the region of H/C>1.0. Following micro-aeration treatment, the distribution range of DOM molecules shared by D0 and D20 did not change significantly, but the number of these shared molecules decreased. The unique DOM molecules in D0 increased, indicating that micro-aeration promoted the release or generation of organic matter in D0. In contrast, the unique DOM molecules in D20 significantly decreased, possibly due to enhanced microbial activity in D20 after micro-aeration, accelerating the consumption or transformation of DOM molecules.

3.5. Microbial community structure

The process of turning large molecules into methane relied on cooperation among different microbial communities. Therefore, studying the changes in microbial species composition is helpful for understanding the inner mechanism. From the bacterial community composition of D0 and D20 sludge samples at the genus level, despite similarities in bacterial composition between them, their relative abundances varied (Fig. S5a). *Bacteroidota*, *Firmicutes*, *Actinobacteria*, and *Chloroflexi* were dominant phyla in each sample. Among them, *Bacteroidota* and *Firmicutes* were two typical hydrolytic-acidification bacteria, playing a significant role in the hydrolysis of organic matter and acidogenesis during AD. *Bacteroidota* was reported to have the ability to hydrolyze polysaccharides and proteins [77], while *Firmicutes* could secrete extracellular enzymes closely related to the metabolism of proteins and lipids [78]. During micro-aeration assisted AD, the relative abundances of *Bacteroidota* and *Firmicutes* in digester D0 both increased. *Firmicutes* were also associated with the conversion of polysaccharides to butyric acid, propionic acid, and acetic acid [79], possessing thick cell walls and the ability to produce endospores to survive under extreme conditions [80]. The higher relative abundance of *Bacteroidota* and *Firmicutes* in D0 after micro-aeration indicated their strong ability to degrade complex substrates, thus leading to a higher hydrolysis rate. In contrast to the control group, the changes in *Bacteroidota* in digester D20 were opposite to *Firmicutes*. *Bacteroidota* and *Firmicutes* had similar growth conditions (substrates, optimum temperature, and pH value), so they might have competed with each other for the degradation of polysaccharides and proteins in the digester. The abundance of *Actinobacteria,* which was associated with humus formation and organic matter degradation, decreased with the introduction of micro-aeration [81]. The introduction of micro-aeration might have had a certain inhibitory effect on *Actinobacteria*, showing its poorer oxygen-tolerance. *Chloroflexi*, reported to be one of the important glucose-degrading microbial communities responsible for acid production and H_2 oxidation homoacetogenesis in the AD process [82], could utilize various carbohydrates and amino acids as substrates [78]. The increase in their relative abundance in the carrier added group confirmed the increase in α-glucosidase activity under micro-aerobic conditions (Fig. 2), which could enhance the hydrolysis of polysaccharides and other complex organic substrates. Additionally, significant enrichment of *Tenericutes* and *Candidatus_Cloacimonetes* was found in digester D20 compared to D0, indicating that the utilization of carrier and micro-aeration enriched the microbial community structure, enhanced the utilization and transformation of organic matters, which ensured the system stability in D20 group.

At the genus level, notable disparities existed in the archaeal community structure between the carrier and control groups (Fig. S5b). The dominant archaea in the control and carrier groups were *Methanothrix* and *Methanosarcina*. *Methanothrix* is an obligate acetatoclastic methanogen, while *Methanosarcina* is a mixotrophic methanogen that can utilize acetate or H_2 and CO_2 as substrates for methane production [83]. Before and after micro-aeration, *Methanothrix* remained relatively stable in the control group, indicating that *Methanothrix* had certain adaptability to oxygen. In the presence of trace amount of oxygen, H_2 tends be oxidized by aerobic respiration to ensure the thermodynamic feasibility of syntrophic fatty acid oxidation [43], which insures the conversion of VFAs to acetate. This explained why *Methanothrix* has a relatively high abundance with less acetate accumulation. After introducing micro-aeration, the relative abundance of *Methanosarcina* in the D20 system decreased by 27.40 %. The decrease in *Methanosarcina* was accompanied by the enrichment of *Methanothrix*, with its relative abundance increasing from 1.58 % to 11.99 %. *Methanoculleus* is an obligate hydrogenotrophic methanogen that can use H_2/CO_2 and formate as carbon sources to produce CH_4. It was enriched in the D20 system after micro-aeration, suggesting that micro-aeration was beneficial to the growth of *Methanoculleus*. *Methanobacterium* is another methanogen utilizing H_2 present in all samples but with a relatively low abundance of approximately 1.50 % at D0. Following micro-aeration treatment, the abundance of *Methanobacterium* further decreased, indicating that micro-aeration inhibited its growth [84]. Compared to D0, the relative abundance of *Methanobacterium* increased at D20 after micro-aeration, which may be a result of the carrier alleviating the toxic effects of oxygen. Overall, compared to the control group, a good balance of two methane production pathways was achieved at D20.

3.6. Changes in key metabolic pathways

Above microbial analysis revealed that carriers and micro-aeration posed significantly influence on the microbial communities with distinct functions in the digestive system. To gain deeper insights into the impact of carriers, micro-aeration, and their combination on microbial metabolic functions and to explore the enhancement mechanism of methane production, the Kyoto Encyclopedia of Genes and Genomes (KEGG) database was used to annotate metabolic pathways before and after micro-aeration treatment (Fig. S6, Table S3). The results show that the relative abundance of oxidative phosphorylation, pyruvate metabolism, butyrate metabolism, and methane metabolism pathways were found to be significantly up-regulated under the effect of micro-aeration. Considering the key roles of specific functional enzymes in metabolism, enzymes expressions in these pathways were analyzed to understand the function mechanism of the micro-aeration with or without carriers.

3.6.1. Oxidative phosphorylation

It is well known that oxidative phosphorylation is one of the basic pathways for ATP biosynthesis. Through micro-aeration treatment, pathways related to oxidative phosphorylation in D0 and D20 showed significant enhancement, with higher relative abundances of carrier groups compared to the control group (Fig. S6, Table S4). This suggested that micro-aeration enhanced ATP biosynthesis, crucial for cellular metabolism and various biochemical reactions requiring energy. As illustrated in Fig. 6a, oxidative phosphorylation synthesis of ATP is composed of a series of embedded membrane enzyme complexes as electron carriers. Notably, except for complex III, the other four parts got improved after micro-aeration treatment (Fig. 6b). Complexes I, III, and IV facilitated the transfer of protons from the matrix to the outer side of the inner membrane, generating a proton concentration gradient essential for driving ATP synthesis and electron transfer to subsequent electron transporter. After micro-aeration treatment, the relative abundances of NADH dehydrogenase nuoEFHJ and nuoCD in complex I in the carrier added digester were significantly higher than those in the control group, catalyzing the oxidation of NADH, transferring electrons to coenzyme Q, and pumping H^+ out of the inner membrane. Complex II also posed a significant increase in the relative abundances of sdhABCD and frdABCD, which promoted the oxidation of succinic acid. Complex III: the cytochrome *c* reductase complex UQCRFS1, facilitated the transfer of electrons from CoQ to cytochrome *c* and simultaneously pumped protons out of the inner membrane, contributing to the proton gradient. Complex IV is a cytochrome *c* oxidase complex, the relative abundances of COX15 and coxB got markedly improved after micro-aeration, catalyzing the reduction of O_2 to form H_2O and releasing H^+ in the intermembrane space. Complex V: a pivotal step in ATP synthesis, showed increased relative abundances of the key functional genes after micro-aeration, including ATP synthases (ATPF1ABEDG, ATPF0ABC, ppa, ppaX, ppk1, and ppk2), indicating the enhancement in ATP synthesis (Fig. 6b). The increase in ATP production could enhance the

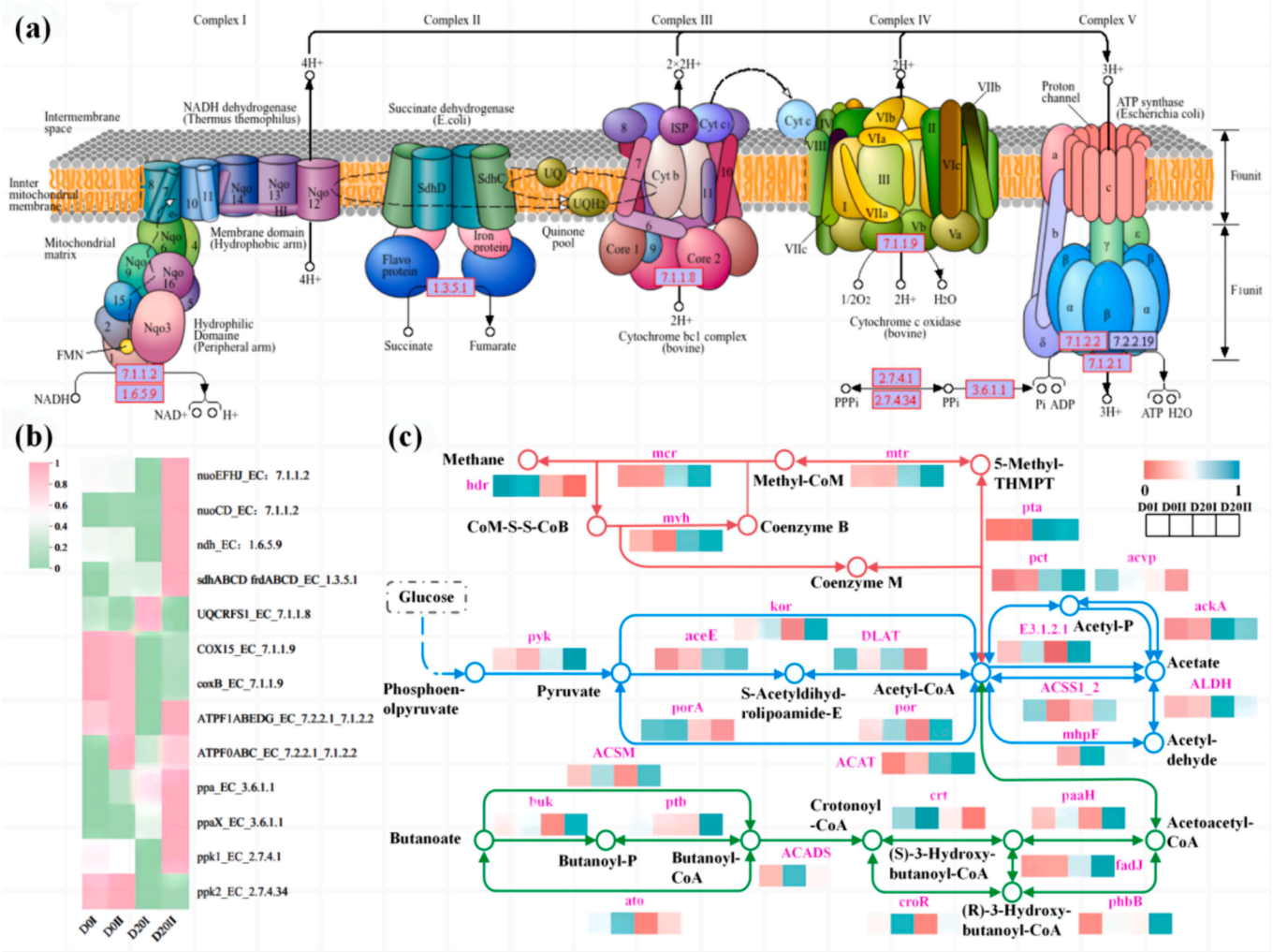


Fig. 6. Relative abundance of key enzymes before and after micro-aeration treatment. (Abundance of enzyme (a) and functional gene (b) in oxidative phosphorylation metabolic pathway; (c) Changes in relative abundance of key metabolic pathways and related enzymes for substrate and methanogenesis.).

anaerobic fermentation process in microorganisms, potentially leading to the higher methane production.

3.6.2. Vfas metabolism

All life activities are energy consuming reactions, and a high level of energy supply may contribute to the conversion of organic matter and the generation of methane. The metabolism of pyruvate is the main pathway for VFA transformation, with acetic acid as the final product. The expression of this metabolic process and related genes is shown in Fig. 6c. Glucose was initially broken down into phosphoenolpyruvate. Pyk is essential in glycolysis and encodes a pyruvate kinase. After micro-aerobic introduction, the expression of pyk in the carrier group was upregulated, 1.5 times higher than in the control group. This upregulation of pyk facilitated the conversion of phosphoenolpyruvate into pyruvate, providing more substrate for pyruvate metabolism to generate acetic acid. As shown in Fig. 6c, pyruvate was converted into acetic acid and acetyl-CoA, and the relative abundance of enzymes involved in the pyruvate metabolism pathway showed a significant upward trend after micro-aeration treatment. The relative abundance of the aceE gene, encoding pyruvate dehydrogenase, was higher in the carrier added group than the control group, and further increased after introduction of micro-aeration, possibly promoting the conversion of pyruvate to acetyl-CoA. DLAT is the functional gene of dihydrolipoamide acetyltransferase, which accepted the acetyl group formed by the oxidative decarboxylation of pyruvate and transferred it to CoA. aceE and DLAT cooperated to convert pyruvate into acetyl-CoA, participating in subsequent methane metabolism. Por and Kor are pyruvate ferredoxin oxidoreductases, which directly converted pyruvate into acetyl-CoA. AckA is a phosphotransferase, and its increased abundance enhanced the yield of acetate. In butyric acid metabolism, genes such as ato, ptb, buk, and ACSM were significantly upregulated in the carrier group after micro-aeration, indicating that the combination of carrier with micro-aeration enhanced the conversion of butyric acid to acetyl-CoA. phbB and paaH were genes encoding acetyl-CoA acyltransferases, whose relative abundance in the carrier group and control group significantly increased by the micro-aeration, 1.4 times and 2.1 times higher than the micro-aeration absent stage (Table S5). ACAT is an acetyl-CoA acyltransferase, whose abundance was also significantly increased by the micro-aeration, far exceeding that of the control group.

In addition to the above pathways, the relative abundance of fatty acid metabolism, methane metabolism and ABC transporter proteins were improved by the micro-aeration. ABC transporter proteins are present on the cell membrane and transport substances inside and outside the cell membrane through synergistic action with ATPase [85]. ABC transporter proteins have numerous substrates, involving nutrients and signal transduction molecules represented by vitamins, amino acids and metal ions [86]. Changes in the ABC transporter reflect substrate utilization. The abundance of methane metabolism increased by 9.24 % and 5.13 %, and the abundance of fatty acid metabolism increased by 6.89 % and 6.45 % in the D20 group and D0 group. The enhancement of fatty acid metabolism may facilitate the hydrolysis of fatty into VFAs and provide available substances for the subsequent methane metabolism. Methane metabolism is a crucial process in methane production, the D20 group showed a significant increase in the abundance of genes encoding key enzymes compared to D0, especially pta, mtr, and mcr. Among them, pta is one of the important enzymes for methane production from acetate metabolism, while mtr and mcr are related to the synthesis of methyl coenzyme A and the reduction of methyl coenzyme. The significant upregulation of these genes after micro-aeration indicated the strengthening of methane production pathways in the D20 group, thereby promoting methane production. Furthermore, the abundance of ABC transporter proteins in both groups increased after micro-aeration. After micro-aeration treatment, the relative abundance of ABC transporter proteins increased by 7.53 % and 11.23 % in the D20 and D0 groups, respectively (Table S3). The upregulation of ABC transporter protein abundance indicates an improved utilization rate of nutrients in the digestive system.

The metabolic state of organisms during the digestion process is also closely associated with DOM. The complex composition of DOM molecules contains valuable molecular information regarding the metabolic state, including reaction substrates, metabolic transformation products, and microbial metabolic by-products [87]. During the AD hydrolysis and acidification stage, complex organic matter undergoes hydrolysis into DOM and subsequent fermentation into VFAs. Notably, pyruvate, butyrate, and acetate metabolism play a crucial role in the acidification of DOM substrates, as evidenced by the significant upregulation of key genes involved in these metabolic pathways following micro-aeration treatment. This upregulation suggests that acidogenic bacteria can more efficiently utilize DOM substrates, leading to the formation of intermediates that promote increased methane production. Moreover, the carrier group demonstrated enhanced metabolic functions after the micro-aeration treatment, which could be attributed to the heightened utilization of DOM and consequently explained its highest methane production.

4. Conclusions

This study aimed to promote the digestive efficiency of FW through combining biological carriers with micro-aeration, and to elucidate the transformation of DOM molecules and the regulation mechanism of microorganisms. The results indicated that an appropriate combination of carriers (filling rate of 20 %) with micro-aeration improved the methane yield from 390.7 mL CH_4/g VS in Stage I to 417.7 mL CH_4/g VS, which was 1.13 times higher than the control group (368.84 mL CH_4/g·VS). The synergistic effect of micro-aeration and carriers ensured digester stability, created a favorable microenvironment, enhanced microbial metabolic activity, expedited the consumption or conversion of DOM molecules, resulting in a reduction of unique DOM molecules in D20 by 59.71 %. Additionally, this approach facilitated EPS secretion, accelerated biofilm formation on the carriers, and provided protection for methanogens against oxygen stress. Moreover, the combination of micro-aeration and carriers enriched the structure of the microbial community, promoted the expression of key enzymes in metabolic pathways such as acetate metabolism, butyrate metabolism, and oxidative phosphorylation, providing sufficient substrate and energy support for the subsequent methane production metabolism.

CRediT authorship contribution statement

Keke Ding: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Baocun Wu:** Software, Investigation, Formal analysis. **Yi Wang:** Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Linji Xu:** Methodology, Formal analysis, Conceptualization. **Miao Liu:** Resources, Methodology, Conceptualization. **Jinjing Xiang:** Methodology, Formal analysis, Conceptualization. **Yongdong Chen:** Software, Methodology, Formal analysis, Conceptualization. **Li Gu:** Writing – review & editing, Validation, Resources, Project administration, Funding acquisition, Conceptualization. **Jinze Li:** Validation, Software, Conceptualization. **Lin Li:** Validation, Methodology, Formal analysis, Conceptualization. **Qiang He:** Supervision, Methodology, Formal analysis, Conceptualization. **Shaowu Liu:** Resources, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

The authors would like to acknowledge the financial support for this work provided by Natural Science Foundation Project of Chongqing (cstc2022NSCQ-MSX0567), Science and Technology Program of Chongqing University (2022CDJYGRH-008).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2024.155731.

References

[1] O. Parthiba Karthikeyan, E. Trably, S. Mehariya, N. Bernet, J.W.C. Wong, H. Carrere, Pretreatment of food waste for methane and hydrogen recovery: A review, Bioresour. Technol. 249 (2018) 1025–1039.
[2] G. Capson-Tojo, M. Rouez, M. Crest, J.-P. Steyer, J.-P. Delgenès, R. Escudié, Food waste valorization via anaerobic processes: a review, Reviews in Environmental Science and Bio/technology. 15 (3) (2016) 499–547.
[3] L. Zhu, B. Wu, Y. Liu, J. Zhang, R. Deng, L. Gu, Strategy to enhance semi-continuous anaerobic digestion of food waste by combined use of calcium peroxide and magnetite, Water Res. 221 (2022) 118801.
[4] Z. Zhao, Y. Zhang, Y. Li, X. Quan, Z. Zhao, Comparing the mechanisms of ZVI and Fe(3)O(4) for promoting waste-activated sludge digestion, Water Res. 144 (2018) 126–133.
[5] S.R. Periyavaram, L. Uppala, P.H.P. Reddy, Hydrothermal carbonization of food waste: Process parameters optimization and biomethane potential evaluation of process water, J. Environ. Manage. 347 (2023) 119132.
[6] Y. Xu, Y. Lu, L. Zheng, Z. Wang, X. Dai, Perspective on enhancing the anaerobic digestion of waste activated sludge, J. Hazard. Mater. 389 (2020) 121847.
[7] S.-F. Fu, K.-Q. Chen, W.-X. Sun, R. Zhu, Y. Zheng, H. Zou, Improved methane production of corn straw by the stimulation of calcium peroxide, Energ. Conver. Manage. 164 (2018) 36–41.
[8] S.-F. Fu, K.-Q. Chen, R. Zhu, W.-X. Sun, H. Zou, R.-B. Guo, Improved anaerobic digestion performance of Miscanthus floridulus by different pretreatment methods and preliminary economic analysis, Energ. Conver. Manage. 159 (2018) 121–128.
[9] N. Yu, B. Guo, Y. Zhang, L. Zhang, Y. Zhou, Y. Liu, Different micro-aeration rates facilitate production of different end-products from source-diverted blackwater, Water Res. 177 (2020) 115783.
[10] D. Ruan, Z. Zhou, H. Pang, J. Yao, G. Chen, Z. Qiu, Enhancing methane production of anaerobic sludge digestion by microaeration: Enzyme activity stimulation, semi-continuous reactor validation and microbial community analysis, Bioresour. Technol. 289 (2019) 121643.
[11] S.-F. Fu, F. Wang, X.-S. Shi, R.-B. Guo, Impacts of microaeration on the anaerobic digestion of corn straw and the microbial community structure, Chem. Eng. J. 287 (2016) 523–528.
[12] S. Xu, A. Selvam, J.W.C. Wong, Optimization of micro-aeration intensity in acidogenic reactor of a two-phase anaerobic digester treating food waste, Waste Manag. 34 (2) (2014) 363–369.
[13] Z. Guo, L. Gao, L. Wang, W. Liu, A. Wang, Enhanced methane recovery and exoelectrogen-methanogen evolution from low-strength wastewater in an up-flow biofilm reactor with conductive granular graphite fillers, Front. Environ. Sci. Eng. 12 (4) (2018) 13.
[14] Y. Chen, J.J. Cheng, K.S. Creamer, Inhibition of anaerobic digestion process: A review, Bioresour. Technol. 99 (10) (2008) 4044–4064.
[15] R.D.A. Cayetano, G.B. Kim, J. Park, Y.H. Yang, B.H. Jeon, M. Jang, S.H. Kim, Biofilm formation as a method of improved treatment during anaerobic digestion of organic matter for biogas recovery, Bioresour. Technol. 344 (Pt B) (2022) 126309.
[16] M.E. Casas, R.K. Chhetri, G. Ooi, K.M. Hansen, K. Litty, M. Christensson, C. Kragelund, H.R. Andersen, K. Bester, Biodegradation of pharmaceuticals in hospital wastewater by staged Moving Bed Biofilm Reactors (MBBR), Water Res. 83 (2015) 293–302.
[17] H.-Y. Jin, Z.-W. He, Y.-X. Ren, C.-C. Tang, A.-J. Zhou, W. Liu, B. Liang, Z.-H. Li, A. Wang, Current advances and challenges for direct interspecies electron transfer in anaerobic digestion of waste activated sludge, Chem. Eng. J. 450 (2022).
[18] T. Wang, L. Shao, T. Li, F. Lu, P. He, Digestion and dewatering characteristics of waste activated sludge treated by an anaerobic biofilm system, Bioresour. Technol. 153 (2014) 131–136.
[19] C.C. Tang, B.C. Zhang, X.Y. Yao, T. Sangeetha, A.J. Zhou, W. Liu, Y.X. Ren, Z. Li, A. Wang, Z.W. He, Natural zeolite enhances anaerobic digestion of waste activated sludge: Insights into the performance and the role of biofilm, J. Environ. Manage. 345 (2023) 118704.
[20] R. Escudie, R. Cresson, J.P. Delgenes, N. Bernet, Control of start-up and operation of anaerobic biofilm reactors: an overview of 15 years of research, Water Res. 45 (1) (2011) 1–10.
[21] W.-J. Gong, H. Liang, W.-Z. Li, Z.-Z. Wang, Selection and evaluation of biofilm carrier in anaerobic digestion treatment of cattle manure, Energy 36 (5) (2011) 3572–3578.
[22] J. Shi, Y. Han, C. Xu, H. Han, Enhanced biodegradation of coal gasification wastewater with anaerobic biofilm on polyurethane (PU), powdered activated carbon (PAC), and biochar, Bioresour. Technol. 289 (2019) 121487.
[23] W. Wang, H. Hou, S. Hu, X. Gao, Performance and stability improvements in anaerobic digestion of thermally hydrolyzed municipal biowaste by a biofilm system, Bioresour. Technol. 101 (6) (2010) 1715–1721.
[24] J. Shi, J. Wang, N. Wan, J. Qian, S. Li, L. Xu, C. Wang, H. Han, Treatment of phenols and NH3–N with activated carbon and polyurethane carrier in biological enrichment reactor, J. Clean. Prod. 369 (2022).
[25] T.A. Kotsopoulos, X. Karamanlis, D. Dotas, G.G. Martzopoulos, The impact of different natural zeolite concentrations on the methane production in thermophilic anaerobic digestion of pig waste, Biosyst. Eng. 99 (1) (2008) 105–111.
[26] S. Fu, S. Lian, I. Angelidaki, R. Guo, Micro-aeration: an attractive strategy to facilitate anaerobic digestion, Trends Biotechnol. 41 (5) (2023) 714–726.
[27] F.-X. Ye, Y.-X. Chen, X.-S. Feng, Advanced start-up of anaerobic attached film expanded bed reactor by pre-aeration of biofilm carrier, Bioresour. Technol. 96 (1) (2005) 115–119.
[28] D. Nguyen, S.K. Khanal, A little breath of fresh air into an anaerobic system: How microaeration facilitates anaerobic digestion process, Biotechnol. Adv. 36 (7) (2018) 1971–1983.
[29] L. Appels, J. Baeyens, J. Degrève, R. Dewil, Principles and potential of the anaerobic digestion of waste-activated sludge, Prog. Energy Combust. Sci. 34 (6) (2008) 755–781.
[30] Y. Xu, Y. Lu, X. Dai, B. Dong, The influence of organic-binding metals on the biogas conversion of sewage sludge, Water Res. Article. 126 (2017) 329–341.
[31] X. Dai, Y. Xu, B. Dong, Effect of the micron-sized silica particles (MSSP) on biogas conversion of sewage sludge, Water Res. Article. 115 (2017) 220–228.
[32] X. Wang, Y. Li, J. Liu, N.Q. Ren, J. Qu, Augmentation of protein-derived acetic acid production by heat-alkaline-induced changes in protein structure and conformation, Water Res. Article. 88 (2016) 595–603.
[33] S. Hao, S. Ren, N. Zhou, H. Chen, M. Usman, C. He, Q. Shi, G. Luo, S. Zhang, Molecular composition of hydrothermal liquefaction wastewater from sewage sludge and its transformation during anaerobic digestion, J. Hazard. Mater. 383 (2020) 121163.
[34] W. Li, Y. Liu, B. Wu, L. Gu, R. Deng, Upgrade the high-load anaerobic digestion and relieve acid stress through the strategy of side-stream micro-aeration: biochemical performances, microbial response and intrinsic mechanisms, Water Res. 221 (2022) 118850.
[35] M. Gao, J. Yang, S. Li, S. Liu, X. Xu, F. Liu, L. Gu, Effects of incineration leachate on anaerobic digestion of excess sludge and the related mechanisms, J. Environ. Manage. 311 (2022) 114831.
[36] X. Peng, S. Zhang, L. Li, X. Zhao, Y. Ma, D. Shi, Long-term high-solids anaerobic digestion of food waste: Effects of ammonia on process performance and microbial community, Bioresour. Technol. 262 (2018) 148–158.
[37] W. Tian, J. Li, L. Zhu, W. Li, L. He, L. Gu, R. Deng, D. Shi, H. Chai, M. Gao, Insights of enhancing methane production under high-solid anaerobic digestion of wheat straw by calcium peroxide pretreatment and zero valent iron addition, Renew. Energy 177 (2021) 1321–1332.
[38] S. Li, Y. Zhang, W. Duan, R. Deng, L. Gu, D. Shi, Insights into the resistance of different extracellular polymeric substance (EPS) layers to the fermentation environment in sludge anaerobic digestion, Chem. Eng. J. 449 (2022).
[39] X. Lu, C. Zheng, G. Zhen, Y. Tan, Y. Zhou, Z. Zhang, C. Niu, W. Li, D. Kudisi, Y. Wang, et al., Roles of colloidal particles and soluble biopolymers in long-term performance and fouling behaviors of submerged anaerobic membrane bioreactor treating methanolic wastewater, J. Clean. Prod. 290 (2021).
[40] M.Y. Chen, D.J. Lee, J.H. Tay, Distribution of extracellular polymeric substances in aerobic granules, Appl. Microbiol. Biotechnol. 73 (6) (2007) 1463–1469.
[41] B. Zhang, C. Shan, Z. Hao, J. Liu, B. Wu, B. Pan, Transformation of dissolved organic matter during full-scale treatment of integrated chemical wastewater: Molecular composition correlated with spectral indexes and acute toxicity, Water Res. 157 (2019) 472–482.
[42] H. Huang, X. Fan, C. Peng, J. Geng, L. Ding, K. Xu, Y. Zhang, H. Ren, Biofilm aging in full-scale aerobic bioreactors from perspectives of metabolic activity and microbial community, Biochem. Eng. J. 146 (2019) 69–78.
[43] Z. Zhou, Q. Ming, Y. An, D. Ruan, G. Chen, H. Wei, M. Wang, Z. Wu, Performance and microbial community analysis of anaerobic sludge digestion enhanced by in-situ microaeration, J. Water Process Eng. 42 (2021).
[44] M. Khademian, J.A. Imlay, How Microbes Evolved to Tolerate Oxygen, Trends Microbiol. 29 (5) (2021) 428–440.
[45] J. Zhao, Y. Li, G.J.W. Euverink, Effect of bioaugmentation combined with activated charcoal on the mitigation of volatile fatty acids inhibition during anaerobic digestion, Chem. Eng. J. 428 (2022).
[46] D. Nguyen, Z. Wu, S. Shrestha, P.H. Lee, L. Raskin, S.K. Khanal, Intermittent micro-aeration: New strategy to control volatile fatty acid accumulation in high organic loading anaerobic digestion, Water Res. 166 (2019) 115080.
[47] K. Wang, Z. Zhou, Y. Zheng, J. Jiang, J. Huang, J. Qiang, Y. An, L. Jiang, L.-M. Jiang, Z. Wang, Understanding mechanisms of sludge in situ reduction in anaerobic side-stream reactor coupled membrane bioreactors packed with carriers at different filling fractions, Bioresour. Technol. 316 (2020).
[48] Y. Zheng, C. Cheng, Z. Zhou, H. Pang, L. Chen, L.-M. Jiang, Insight into the roles of packing carriers and ultrasonication in anaerobic side-stream reactor coupled membrane bioreactors: Sludge reduction performance and mechanism, Water Res. 155 (2019) 310–319.

[49] Z. Zhang, Y. Yu, H. Xi, Y. Zhou, Review of micro-aeration hydrolysis acidification for the pretreatment of toxic and refractory organic wastewater, J. Clean. Prod. 317 (2021).
[50] G.Z. Kyzas, G. Bomis, R.I. Kosheleva, E.K. Efthimiadou, E.P. Favvas, M. Kostoglou, A.C. Mitropoulos, Nanobubbles effect on heavy metal ions adsorption by activated carbon, Chem. Eng. J. 356 (2019) 91–97.
[51] X. Ye, Z. Jia, Y. Liu, C. Wang, C. Cao, Y. Zhang, T. Han, L. Wang, T. Guo, Y. Xi, Constructing carbon-based materials loaded with MOFs to realize efficient anaerobic digestion of rural organic waste, Fuel 355 (2024).
[52] J. Yang, J. Zhang, C. Liu, J. Liu, L. Zhang, Performance of a fixed-bed biofilm reactor with microbubble aeration in aerobic wastewater treatment, Water Sci. Technol. 74 (1) (2016) 138–146.
[53] H.-C. Flemming, J. Wingender, Relevance of microbial extracellular polymeric substances (EPSs) - Part I: Structural and ecological aspects, Water Sci. Technol. 43 (6) (2001) 1–8.
[54] J. Wingender, T.R. Neu, H.-C. Flemming, Microbial extracellular polymeric substances: characterization, structure, and function, Springer, 1999.
[55] X. Guo, X. Wang, J. Liu, Composition analysis of fractions of extracellular polymeric substances from an activated sludge culture and identification of dominant forces affecting microbial aggregation, Sci. Rep. 6 (1) (2016) 28391.
[56] C.S. Laspidou, B.E. Rittmann, A unified theory for extracellular polymeric substances, soluble microbial products, and active and inert biomass, Water Res. 36 (11) (2002) 2711–2720.
[57] Y. Zhang, H. Ma, Q. Niu, R. Chen, T. Hojo, Y.-Y. Li, Effects of substrate shock on extracellular polymeric substance (EPS) excretion and characteristics of attached biofilm anammox granules, RSC Adv. 6 (114) (2016) 113289–113297.
[58] X. Tian, Z. Shen, Z. Han, Y. Zhou, The effect of extracellular polymeric substances on exogenous highly toxic compounds in biological wastewater treatment: An overview, Bioresource Technology Reports. 5 (2019) 28–42.
[59] L. Zhang, X. Feng, N. Zhu, J. Chen, Role of extracellular protein in the formation and stability of aerobic granules, Enzyme Microb. Technol. 41 (5) (2007) 551–557.
[60] D. Lu, K. Xiao, Y. Chen, Y.N.A. Soh, Y. Zhou, Transformation of dissolved organic matters produced from alkaline-ultrasonic sludge pretreatment in anaerobic digestion: From macro to micro, Water Res. 142 (2018) 138–146.
[61] D.E. LaRowe, P. Van Cappellen, Degradation of natural organic matter: A thermodynamic analysis, Geochim. Cosmochim. Acta 75 (8) (2011) 2030–2042.
[62] X. Chen, J. Liu, J. Chen, J. Wang, X. Xiao, C. He, Q. Shi, G. Li, N. Jiao, Oxygen availability driven trends in DOM molecular composition and reactivity in a seasonally stratified fjord, Water Res. 220 (2022) 118690.
[63] A.M. Kellerman, D.N. Kothawala, T. Dittmar, L.J. Tranvik, Persistence of dissolved organic matter in lakes related to its molecular characteristics, Nat. Geosci. 8 (6) (2015) 454–457.
[64] T. Reemtsma, Determination of molecular formulas of natural organic matter molecules by (ultra-) high-resolution mass spectrometry: Status and needs, J. Chromatogr. A 1216 (18) (2009) 3687–3701.
[65] B. Zhang, C. Shan, S. Wang, Z. Fang, B. Pan, Unveiling the transformation of dissolved organic matter during ozonation of municipal secondary effluent based on FT-ICR-MS and spectral analysis, Water Res. 188 (2021) 116484.
[66] P. Phungsai, F. Kurisu, I. Kasuga, H. Furumai, Changes in Dissolved Organic Matter Composition and Disinfection Byproduct Precursors in Advanced Drinking Water Treatment Processes, Environ. Sci. Tech. 52 (6) (2018) 3392–3401.
[67] M. Liu, N.J.D. Graham, L. Xu, K. Zhang, W. Yu, Bubbleless Air Shapes Biofilms and Facilitates Natural Organic Matter Transformation in Biological Activated Carbon, Environ. Sci. Tech. 57 (11) (2023) 4543–4555.
[68] S. Kim, R.W. Kramer, P.G. Hatcher, Graphical Method for Analysis of Ultrahigh-Resolution Broadband Mass Spectra of Natural Organic Matter, the Van Krevelen Diagram, Anal. Chem. 75 (20) (2003) 5336–5344.
[69] P. Zhang, P. Huang, X. Xu, H. Sun, B. Jiang, Y. Liao, Spectroscopic and molecular characterization of biochar-derived dissolved organic matter and the associations with soil microbial responses, Sci. Total Environ. 708 (2020) 134619.
[70] W.D. Gnowe, E. Noubissié, G.B. Noumi, Influence of time and oxygenation on the degradation of organic matter, nitrogen and phosphates during the biological treatment of slaughterhouse effluent, Case Stud. Chem. Environ. Eng.. 2 (2020) 100048.
[71] K. Xiao, Z. Yu, K. Pei, M. Sun, Y. Zhu, S. Liang, H. Hou, B. Liu, J. Hu, J. Yang, Anaerobic digestion of sludge by different pretreatments: Changes of amino acids and microbial community, Front. Environ. Sci. Eng. 16 (2) (2021) 23.
[72] S. Shakeri Yekta, A. Elreedy, T. Liu, M. Hedenström, S. Isaksson, M. Fujii, A. Schnürer, Influence of cysteine, serine, sulfate, and sulfide on anaerobic conversion of unsaturated long-chain fatty acid, oleate, to methane, Sci. Total Environ. 817 (2022) 152967.
[73] R. Reshmy, P. Athiyaman Balakumaran, K. Divakar, E. Philip, A. Madhavan, A. Pugazhendhi, R. Sirohi, P. Binod, M. Kumar Awasthi, R. Sindhu, Microbial valorization of lignin: Prospects and challenges, Bioresour. Technol. 344 (2022) 126240.
[74] R. Yuan, Y. Shen, N. Zhu, C. Yin, H. Yuan, X. Dai, Pretreatment-promoted sludge fermentation liquor improves biological nitrogen removal: Molecular insight into the role of dissolved organic matter, Bioresour. Technol. 293 (2019) 122082.
[75] B. Wu, H. Wang, W. Li, X. Dai, X. Chai, Influential mechanism of water occurrence states of waste-activated sludge: Potential linkage between water-holding capacity and molecular compositions of EPS, Water Res. 213 (2022) 118169.
[76] S. Shakeri Yekta, M. Gonsior, P. Schmitt-Kopplin, B.H. Svensson, Characterization of Dissolved Organic Matter in Full Scale Continuous Stirred Tank Biogas Reactors Using Ultrahigh Resolution Mass Spectrometry: A Qualitative Overview, Environ. Sci. Tech. 46 (22) (2012) 12711–12719.
[77] K. Kundu, I. Bergmann, M. Klocke, S. Sharma, T.R. Sreekrishnan, Impact of abrupt temperature increase on the performance of an anaerobic hybrid bioreactor and its intrinsic microbial community, Bioresour. Technol. 168 (2014) 72–79.
[78] H. Zhou, R.C. Brown, Wen, Z, Reducing toxicity and improving microbial tolerance. Bioresource Technology Article, Anaerobic digestion of aqueous phase from pyrolysis of biomass, 2019, p. 292.
[79] Z.-W. He, W.-Z. Liu, C.-C. Tang, B. Liang, Z.-C. Guo, L. Wang, Y.-X. Ren, A.-J. Wang, Performance and microbial community responses of anaerobic digestion of waste activated sludge to residual benzalkonium chlorides, Energ. Conver. Manage. 202 (2019) 112211.
[80] L. Zhou, Y. Gao, K. Yu, H. Zhou, Y.G. De Costa, S. Yi, W.-Q. Zhuang, Microbial community in in-situ waste sludge anaerobic digestion with alkalization for enhancement of nutrient recovery and energy generation, Bioresour. Technol. 295 (2020) 122277.
[81] W. Zhang, G. Zhang, G. Liu, Z. Dong, T. Chen, M. Zhang, P.J. Dyson, L. An, Bacterial diversity and distribution in the southeast edge of the Tengger Desert and their correlation with soil enzyme activities, J. Environ. Sci. 24 (11) (2012) 2004–2011.
[82] H.D. Ariesyady, T. Ito, S. Okabe, Functional bacterial and archaeal community structures of major trophic groups in a full-scale anaerobic sludge digester, Water Res. 41 (7) (2007) 1554–1568.
[83] M.C. Nelson, M. Morrison, Z. Yu, A meta-analysis of the microbial diversity observed in anaerobic digesters, Bioresour. Technol. 102 (4) (2011) 3730–3739.
[84] A. Kiener, T. Leisinger, Oxygen Sensitivity of Methanogenic Bacteria, Syst. Appl. Microbiol. 4 (3) (1983) 305–312.
[85] Y. Wang, J. Li, M. Liu, L. Gu, L. Xu, J. Li, L. Ao, Enhancement of anaerobic digestion of high salinity food waste by magnetite and potassium ions: Digestor performance, microbial and metabolomic analyses, Bioresour. Technol. 388 (2023) 129769.
[86] W. Li, L. Zhu, B. Wu, Y. Liu, J. Li, L. Xu, X. Huangfu, D. Shi, L. Gu, C. Chen, Improving mesophilic anaerobic digestion of food waste by side-stream thermophilic reactor: Activation of methanogenic, key enzymes and metabolism, Water Res. 241 (6) (2023) 120167–120179.
[87] J. Liu, C. Wang, Z. Hao, G. Kondo, M. Fujii, Q.L. Fu, Y. Wei, Comprehensive understanding of DOM reactivity in anaerobic fermentation of persulfate-pretreated sewage sludge via FT-ICR mass spectrometry and reactomics analysis, Water Res. 229 (2023) 119488.